TARUN RAMPRAKASH TA
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Computational Engineer | Derby, UK

PERSONAL STATEMENT

Computational engineer with an MSc in Aerospace Engineering and more than 2-year experience in mathematical modelling, numerical algorithms and CFD across academic research and industrial product development within UK and India. Strong background in Python/C++ with hands on experience in AI/ML techniques, parallel computing and validation studies. Experienced in modelling and simulation tools and skilled in translating research-grade algorithms into validated engineering tools. Recognised for simplifying complex engineering problems into robust computational solutions while collaborating effectively with software developers, domain specialists and research teams.

KEY ACHIEVEMENTS

- Developed a Viscous Full Potential flow solver in Python that modernized a legacy aerodynamic analysis workflow, enabling rapid conceptual aircraft design and supporting Airbus OneHeart Design Sprint activities (Cranfield University, 2026)
- Led a multidisciplinary engineering team of five members delivering the Air Orca UAV concept, coordinating aerodynamic, structural and manufacturing activities from concept through validation (BONV Technologies, India, 2024)
- Authored three international conference publications in computational aerodynamics and numerical modelling, demonstrating research contributions across high-fidelity simulation, aerodynamic optimisation and solver development (2024–2026)

EDUCATION

MSc by Research in Aerospace Engineering, Cranfield University, UK **Sept 2024 - April 2026**

- **Research Thesis:** Development of Viscous Full Potential (VFP) Transonic Flow Solver for Rapid Conceptual Aircraft Design Analysis
- Designed and implemented mathematical model of Propeller Slipstream based on actuator disk theory and tailplane Aerodynamics improving the fidelity of legacy workflows and widening the scope for future AI/ML integration
- **Tech Stacks:** - Mathematical Modelling, Numerical Methods, Python, MATLAB, Git, CFD

B. Tech in Materials Engineering, Indian Institute of Technology (IIT), Varanasi, India **Nov 2020 - May 2024**

- **Modules:** CS101 (C++17 and Python), Engineering Mathematics, Numerical Techniques, Partial Differential Equations, Calculus, Deformation and Fracture, Strength of Materials, Composite Materials, Material Testing

PROFESSIONAL EXPERIENCES

FlowThermo Lab, Nottingham, UK, CFD Intern **Jan 2026 – Aug 2026**
A global computational engineering consultancy specialising in CFD, FEA and simulation-driven product development.

- Evaluated and benchmarked three gradient reconstruction algorithms (Green-Gauss and Least Squares) in C++ to quantify numerical accuracy, convergence behaviour and solver stability across representative CFD cases.

- Developed finite-volume and finite-difference numerical algorithms in C++ and MATLAB, implementing a parallel 2D Euler solver using MPI to validate shock-capturing performance against benchmark problems.
- Investigated turbulence model sensitivity and numerical method behaviour across study cases in Fluent and OpenFOAM, applying CFD best practices to refine the modelling approach and strengthen the reliability of simulation.

BONV Technologies, India, Research and Development Engineer **Dec 2023 - Sep 2024**

Established in 2021, BONV is an Indian aerospace company specialising in eVTOLs and high-payload UAVs.

- Translated mission-level performance requirements into aerodynamic, structural and manufacturing design solutions, balancing multidisciplinary trade-offs across multiple UAV development programmes.
- Performed engineering trade-off studies across more than 25 designs to optimise UAV configurations, integrating aerodynamic performance, payload capability and manufacturing constraints into final design decisions.
- Coordinated aerodynamics and flight dynamics requirements across a multi-disciplinary engineering team, translating operational performance targets into design specifications.

INTERNSHIP EXPERIENCE

Cambridge University, Remote, Research Intern **Nov 2023 - Jan 2024**

- Created a physics-informed neural operator (PINO) model using PyTorch for estimating the friction factor responsible for indirect noise in aircraft non-isentropic nozzles, providing estimates with an uncertainty of 2-3%.
- Developed a physics-informed neural network to solve non-linear PDEs with a focus on fluid dynamics applications, attaining an accuracy of 95% in simulating Navier-Stokes equations for flows past a cylinder.

Cranfield University, Cranfield, UK, Research Intern **May 2023 - Aug 2023**

- Performed CFD validation studies in ANSYS Fluent and FlightStream, comparing mid-fidelity vortex methods to RANS CFD and wind-tunnel measurements to evaluate aerodynamic efficiency gains and supporting publication at ICAS 2024.
- Led and executed wind tunnel test campaigns at the National Wind Tunnel Facility on leading-edge tubercle wings, including test matrix design, model preparation checks and data-acquisition validation to ensure repeatable measurements.

SKILLS, INTERESTS AND EXTRACURRICULAR ACTIVITIES

- **Languages:** Fluent in English, Hindi and Tamil
- **IT Skills:** Confident IT user, with proficiency in MS Office workflows
- **Modelling/Simulation –** Proven experience in MALTAB, SolidWorks, ANSYS and OpenFOAM
- **Programming/Software:** Python, C/C++ for numerical frameworks and software development with Git for version control. Bash/Shell scripting for automation & file handling on Linux/HPC and remote server workflows
- **Individual Interests:** Powered by football, cricket, coffee, and the occasional afternoon spent plane spotting
- **Professional/Technical Training:** AI/ML for Fluid Mechanics (FlowThermo Lab, 2026), Machine Learning (Stanford Online, 2022)